2025.12.23 HW12: 习题七.

A11. (1) $\overline{x} = \frac{22.3+21.5+21.8+21.4+22.1}{5} = 21.82$

由 $\sigma = 0.3$, $z_{0.025} = 1.96$.

$1.96 \times \frac{0.3}{\sqrt{5}} = 0.26296$.

$\therefore$ 置信区间为 $(21.557, 22.083)$

(2) $2 \times 1.96 \times \frac{0.3}{\sqrt{n}} \leq 0.2 \Rightarrow n \geq 35$

A12. (1) $\overline{x} = \frac{4040+2990+2964+3245+3626+3633+3387+4136+3595+3194+3714+2831+3845+3410+3004}{15}$

$= 3400.93$.

$S = 412.79$

$t_{0.025}(14) = 2.1448$

$\therefore 2.1448 \times \frac{412.79}{\sqrt{15}} = 228.597$, 故置信区间为 $(3172.3336, 3629.5330)$.

(2) $t_{0.05}(14) = 1.7613$.

$1.7613 \times \frac{412.79}{\sqrt{15}} = 187.7227$.

$\therefore$ 单侧下限为 $3400.93 - 187.7227 = 3213.2073$.

A14. d: 2, -7, -13, -18, -5, -11, -17, 4, -9, 7

$\bar{d}=-6.7$, $S_d=8.6801$

$t_{0.025}(9)=2.262$.

∴ 边际误差 $2.262\times\frac{8.6801}{\sqrt{10}}=6.208$

置信区间 $(-12.9095, -0.4905)$

A.16 (1) $\chi^2_{0.025}(15)=27.488$, $\chi^2_{0.975}(15)=6.262$

$(n-1)S^2=(16-1)\times 300^2=1350000$

σ^2 下限 $\frac{1350000}{27.488}=49112.78$

σ^2 上限 $\frac{1350000}{6.262}=215585.44$

$\therefore \sigma\in(221.61, 464.31)$

(2) $\chi^2_{0.95}(15)=7.261$

$\frac{1350000}{7.261}=185924.25$, $U=\sqrt{185924.25}=431.19$

A18. (1) $\bar{x}-\bar{y}=456.64-451.34=5.3$.

$$\sqrt{\frac{13^2}{16}+\frac{18^2}{12}}=4.75.$$

$z_{0.025}=1.96$，边际误差 $1.96\times4.75=9.31$.

∴置信区间为 $(-4.01, 14.61)$.

(2) $$S_p^2=\frac{15\times12.8^2+11\times11.3^2}{26}=148.5465.$$

∴ 标准误差 $\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}=4.654$.

$t_{0.025}(26)=2.056$.

边际误差 $2.056\times4.654=9.57$.

∴置信区间为 $(-4.27, 14.87)$

(3) $S_1^2=163.84$ $\quad F_{0.025}(15,11)=3.33$

$S_2^2=127.69$ $\quad F_{0.025}(11,15)=3.007$.

$$\frac{S_1^2}{S_2^2}=1.283.$$

∴下限 $=\frac{1.283}{3.33}=0.385$，上限 $=1.283\times3.007=3.858$.

∴ $\frac{\sigma_1^2}{\sigma_2^2}$ 的95%置信区间为 $(0.385, 3.858)$.